

Distribution-independent SQ learning does not imply low dimension complexity

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Abstract

Let $\text{rect}(H)$ be the minimum, over product distributions on examples and hypotheses, of the largest monochromatic-rectangle mass. We give one distribution-independent randomized statistical-query learner for every finite class with $\text{rect}(H) \geq 1/\rho$. For every $0 < \epsilon < 1/4$ it uses $O(\rho(\log |H| + \log(1/\epsilon)))$ queries of tolerance $\Omega(\epsilon/(\rho \log(1/\epsilon)))$, and achieves expected error at most ϵ against every valid adaptive oracle. The algorithm solves finite rectangle-mixture linear programs; its computation need not be efficient. Applied to the random monotone incidence flips of Hatami, Hatami, Pires, Tao, and Zhao, this gives classes on $2M$ points with constant inverse rectangle ratio, $O(\log M)$ queries at constant accuracy, and dimension $\Omega(M^{1/3}/\log M)$. Exact probabilistic dimension is $\Omega(M^{1/3}/\log^2 M)$; for every fixed $\eta < 1/2$, averaged probabilistic dimension is $\Omega_\eta(M^{1/3}/\log M)$. Thus all three proposed polynomial implications from distribution-independent SQ learning to these dimension notions fail. The class selection is non-explicit; the learner is explicit given its truth table. We prove a randomized statistical-dimension bound, retain the joint-distribution spectral barrier and the projective-plane analysis, and identify what does not transfer to ordinary bias-free ReLU SGD. CPU checks executed. The four additional Lean proof scripts are pinned but uncompiled.

1 Introduction

A fixed linear representation must encode the signs of every example–hypothesis pair at once. Statistical queries can instead adapt to the unknown marginal distribution. The distinction persists even when a single learner must work for every marginal: distribution independence does not prevent distribution adaptation.

We prove that monochromatic rectangles under product distributions suffice for a common SQ learner. The key is a finite minimax step that produces a distribution over rectangles from the current hypothesis prior, without knowing the example distribution. A sampled rectangle either permits safe labeling of examples or eliminates hypotheses. We account for these alternatives through the product of their two masses. This avoids paying the inverse rectangle ratio twice in the query count.

The incidence-flip construction of Hatami et al. [2] separates bounded inverse rectangle ratio from sign-rank. Its geometric template has sign-rank at most six and no all-negative 2×2 submatrix. Arbitrary monotone flips preserve enough monochromatic rectangles, whereas independent flips supply many unconstrained signs. Combining the rectangle learner with the counting argument answers the SQ question of Feldman et al. [1] negatively. This is an information-theoretic statement. Neither polynomial-time evaluation of the learner nor a small ordinary neural network is asserted.

The probabilistic dimension statements require different arguments. Uniform approximation on the whole matrix would miss almost all of the randomly flipped incidences. We instead test

each target under the uniform distribution on its own template incidences. For exact probabilistic dimension, a full-support marginal and a union bound over hypotheses suffice.

We use two classical inputs with their precise constants: the Warren strict-sign-pattern bound, and, only for the sharper numerical rectangle constant 2^{-15} , the homogeneous-rectangle theorem of Alon et al. [3]. We derive their application-specific consequences explicitly. A separate radial-isotropy proof below gives the weaker constant 2^{-18} without the latter geometric input and already proves the qualitative separation. We do not claim to reprove Warren’s general real-algebraic theorem. The new learner and the probabilistic reductions do not invoke an unverified SQ characterization.

2 Definitions and quantifiers

Let X and the nonempty indexed class $H = \{h_1, \dots, h_K\} \subseteq \{-1, +1\}^X$ be finite. Duplicate rows cause no difficulty. Set $n = \log_2 |X|$. For a distribution D on X and Boolean predictor g , define

$$L_{D,h}(g) = \sum_x D(x) \mathbf{1}_{g(x) \neq h(x)}, \quad \mathbb{E}_D[hg] = 1 - 2L_{D,h}(g).$$

A query is any $q : X \times \{-1, +1\} \rightarrow [-1, 1]$. At tolerance $\tau > 0$, a valid oracle returns any number within τ of $\mathbb{E}_D q(x, h(x))$. Its choices may depend on the query history. A randomized learner must satisfy

$$\forall D \forall h \in H \forall \text{ valid oracle policies } O, \quad \mathbb{E}_\omega L_{D,h}(A_\omega^O) \leq \epsilon. \quad (1)$$

The private seed is sampled independently of the instance. A deterministic algorithm is a real-response decision tree; a randomized algorithm is a distribution over such trees. Query count is a worst-case bound. Computation and randomness length are not charged. All random variables used here are measurable; our learner has finitely many discrete states and uses only threshold comparisons of answers.

The dimension $\text{dc}(H)$ is the least $d \geq 0$ admitting $\phi : X \rightarrow \mathbb{R}^d$ such that every h has a vector w_h with

$$h(x) \langle w_h, \phi(x) \rangle > 0 \quad (x \in X). \quad (2)$$

It equals the sign-rank of the truth table, since a rank factorization is exactly such a representation. It is finite by $\phi(x) = e_x$.

For the probabilistic variants we fix $\text{sign}(0) = +1$, without changing the strict definition above. The quantity $\text{dc}_{1/2}(H)$ is the least d for which one distribution over embeddings satisfies

$$\forall D, h, \quad \Pr_\phi[\exists w : L_{D,h}(\text{sign}\langle w, \phi \rangle) = 0] \geq 1/2.$$

The quantity $\text{dc}^\eta(H)$ instead requires

$$\forall D, h, \quad \mathbb{E}_\phi \inf_w L_{D,h}(\text{sign}\langle w, \phi \rangle) \leq \eta.$$

The embedding law is independent of both D and h in each definition.

A monochromatic rectangle is $S \times T \subseteq X \times H$ with $h(x) = c$ for all $x \in S, h \in T$, for some $c \in \{-1, +1\}$. Define

$$\text{rect}(H) = \inf_{\mu \in \Delta(X), \nu \in \Delta(H)} \max_{S \times T \text{ monochromatic}} \mu(S) \nu(T).$$

All rectangles and simplices are finite-dimensional. Restricting to any nonempty submatrix cannot decrease rect , by extending its two distributions by zero. In particular, a known rational $r = 1/\rho \leq \text{rect}(H)$ remains valid throughout the algorithm.

3 The rectangle learner

The earlier barrier and transcript results retain numbers 1–5 in Appendix A. The main new results start at 9.

Theorem 9 (Distribution-independent rectangle learner). *Suppose $\rho \geq 1$ and $\text{rect}(H) \geq 1/\rho$. For $0 < \epsilon < 1/4$, set*

$$r = 1/\rho, \quad P = 1 + \lceil 4\rho \ln(8/\epsilon) \rceil, \quad \tau = \frac{\epsilon}{24P}, \quad (3)$$

$$B = \ln K + \ln(8/\epsilon) + 1, \quad R = \lceil 8\rho(B + \ln(2/\epsilon)) \rceil. \quad (4)$$

There is an explicit randomized learner satisfying (1), with at most $m = 3R + 1$ queries, all at tolerance τ . Thus

$$m = O(\rho(\ln K + \ln(1/\epsilon))), \quad \tau = \Omega\left(\frac{\epsilon}{\rho \ln(1/\epsilon)}\right).$$

The algorithm is explicit from the finite truth table and a lower bound r . It may use exponentially large finite linear programs.

Lemma 10 (A marginal-free rectangle mixture). *Fix nonempty $U \subseteq X$, nonempty $V \subseteq H$, and a distribution ν on V . There is a distribution λ over monochromatic rectangles $S_R \times T_R \subseteq U \times V$ such that*

$$\forall x \in U, \quad \mathbb{E}_{R \sim \lambda}[\mathbf{1}_{S_R}(x)\nu(T_R)] \geq r. \quad (5)$$

This distribution can be computed from (H, U, V, ν) without a query about D .

Proof. For each rectangle let $z_R \in \mathbb{R}^U$ have coordinates $z_R(x) = \mathbf{1}_{S_R}(x)\nu(T_R)$. The rectangle assumption gives

$$\forall \mu \in \Delta(U), \quad \max_R \langle \mu, z_R \rangle \geq r.$$

Let C be the convex hull of the finitely many z_R and let $Q = r\mathbf{1} + \mathbb{R}_{\geq 0}^U$. Suppose $C \cap Q = \emptyset$. The distance between compact C and closed Q has a positive attained minimum: a minimizing sequence has bounded Q -components and convergent subsequences. Let (c, z) attain it and put $v = z - c \neq 0$. Differentiating squared distance along segments to any $c' \in C, z' \in Q$ gives

$$\langle v, c' - c \rangle \leq 0, \quad \langle v, z' - z \rangle \geq 0.$$

Taking $z' = z + te_x$ gives $v_x \geq 0$. Hence

$$\max_{c' \in C} \langle v, c' \rangle \leq \langle v, c \rangle < \langle v, z \rangle = \min_{z' \in Q} \langle v, z' \rangle = r \sum_x v_x.$$

Normalize v to a distribution to contradict the assumption. A point in $C \cap Q$ supplies λ . Enumerating all rectangles and solving these finite linear inequalities computes it. For rational ν, r , the feasible polytope has rational coefficients and admits a rational feasible point: Gaussian elimination on a minimal face yields such a point. Thus no oracle about D enters this computation. $\square$

Algorithm. Initialize $U = X, V = H$, and an empty partial predictor. At every state let ν be uniform on V . Use the constants in (3)–(4). Repeat the following for at most R rounds.

1. If $U = \emptyset$, stop. Query $\mathbf{1}_U(x)$ and call its answer $\hat{u}$. If $\hat{u} \leq \epsilon/4$, stop and fill U with $+1$.

2. Compute the distribution in Lemma 10. Draw a monochromatic rectangle $S \times T$ of color c . Query $\mathbf{1}_S(x)$, with answer $\hat{s}$. If $\hat{s} < (r/2)\hat{u}$, reject this proposal and go to the next round.
3. Query $\mathbf{1}_S(x)\mathbf{1}_{y \neq c}$, with answer $\hat{w}$. If $\hat{w} \leq 2\tau$, label S by c and replace U by $U \setminus S$. Otherwise replace V by $V \setminus T$.
4. At the round limit, stop and fill any remaining points with $+1$.

Every accepted peel removes a subset of the current U , so peeled sets are disjoint. An elimination does not change U . There is no candidate verification test, conditional-label query, isotropic transform, or unsafe subspace restriction.

Proof of Theorem 9. Fix D, h^* and any valid oracle policy. First,

$$\tau \leq \epsilon/8, \quad \tau \leq r\epsilon/64. \quad (6)$$

Indeed $P \geq 4\rho \ln(8/\epsilon) > 4\rho \ln 32$, and $24 \cdot 4 \ln 32 > 64$. At a nonstopping round, write

$$u = D(U) > \epsilon/4 - \tau \geq \epsilon/8, \quad a = D(S)/u, \quad b = \nu(T), \quad w = D(S \cap \{h^* \neq c\}).$$

Thus $\tau/u \leq r/8$. If the proposal is accepted, validity implies

$$D(S) \geq (r/2)u - (1 + r/2)\tau \geq (5r/16)u \geq (r/4)u.$$

Conversely, every rectangle with $a \geq 3r/4$ is accepted, since

$$\hat{s} - (r/2)\hat{u} \geq (r/4)u - (1 + r/2)\tau \geq (r/16)u > 0.$$

These inequalities hold at the extreme valid oracle answers, not only for exact answers.

If $h^* \in T$, monochromaticity gives $w = 0$, hence $\hat{w} \leq \tau < 2\tau$. Therefore the algorithm never eliminates the target. In particular, V remains nonempty. On an elimination, its normalized target weight increases by a factor $1/(1-b)$. Because $-\ln(1-b) \geq b$ for $0 \leq b < 1$, summing gives

$$\sum_{\text{eliminations}} b \leq \ln K. \quad (7)$$

For completeness, $\ln t \leq t-1$ for $t > 0$ follows by minimizing $t-1-\ln t$, whose derivative is $1-1/t$. Taking $t = 1-b$ proves the displayed inequality. The target weight starts at $1/K$ and never exceeds one.

Each peel multiplies u by at most $1-r/4$. Immediately before every peel, $u \geq \epsilon/8$. Before the j th peel,

$$\epsilon/8 \leq (1-r/4)^{j-1} \leq e^{-(j-1)r/4}.$$

Hence there are at most P peels. Their combined wrong-color mass is at most $3P\tau = \epsilon/8$, since a peeling answer implies $w \leq 3\tau$. At a normal stop the residual mass is at most $\epsilon/4 + \tau$, or zero. The error at such a stop is at most $\epsilon/2$.

It remains to bound the probability of reaching the round limit. Define a round's progress by $G = a$ for a peel, $G = b$ for an elimination, and $G = 0$ for a rejected proposal. Conditional on the pre-proposal history, $0 \leq G \leq 1$. For every accepted rectangle, either action has $G \geq ab$. All rectangles with $a \geq 3r/4$ are accepted. Averaging (5) under D restricted to U gives $\mathbb{E}[ab] \geq r$. Consequently,

$$\mathbb{E}[G \mid \text{history}] \geq \mathbb{E}[ab] - \mathbb{E}[ab\mathbf{1}_{a < 3r/4}] \geq r - (3r/4)\mathbb{E}b \geq r/4. \quad (8)$$

This uses no favorable distributional assumption and holds for either oracle-induced action on every accepted rectangle.

On any trajectory of nonstopping rounds, the sum of all peeling a values except the last is at most the logarithm of the reciprocal residual mass immediately before the last peel. Indeed $a \leq -\ln(1-a)$ and the residual ratios telescope. This sum is at most $\ln(8/\epsilon)$; the last a is at most one. If there are no peels the sum is zero. Together with (7), this proves

$$\sum G \leq B. \quad (9)$$

There is no logarithm of zero: a peel with $a = 1$ must be the last peel.

After any normal termination, extend the sequence by setting $G = 1$. Let $p = r/4$. For $0 \leq g \leq 1$, convexity places e^{-g} below the chord between its endpoint values:

$$e^{-g} \leq 1 - (1 - e^{-1})g.$$

Since $1 - e^{-1} > 1/2$, conditional expectation and (8) imply

$$\mathbb{E}[e^{-G} \mid \text{history}] \leq e^{-p/2}.$$

Induction gives $\mathbb{E}e^{-\sum_{t=1}^R G_t} \leq e^{-pR/2}$. On the event of no normal termination by the budget, (9) holds. Markov's inequality therefore gives

$$\Pr[\text{budget termination}] \leq e^{B-pR/2} \leq \epsilon/2.$$

A point that becomes empty after the final peel already has the normal-stop error bound; counting it as budget termination only weakens this estimate. Finally, error is always at most one, so expected error is at most $\epsilon/2 + \epsilon/2 = \epsilon$. There are at most three queries per round and at most one extra stopping check. The big- O and Ω bounds follow directly from the ceilings and $\rho \geq 1$, $\epsilon < 1/4$. $\square$

Domain-size consequence of Theorem 9. The dependence on $\log K$ can always be bounded in terms of the domain and ρ . If H shatters $v \geq 1$ points, put the uniform marginal on those points and the uniform prior on one representative of each of their 2^v label patterns. A nonempty monochromatic rectangle using s of the points has mass at most $s/(v2^s) \leq 1/(2v)$, because $s \leq 2^{s-1}$ for every positive integer s (by induction). Therefore

$$\text{VC}(H) \leq \rho/2.$$

After removing duplicate rows, the finite counting inequality $K \leq \sum_{i=0}^v \binom{|X|}{i}$ follows by induction on $|X|$: split restrictions on the last point into their union A and intersection B . Their sizes sum to K , the VC dimension of A is at most v , and that of B is at most $v-1$, since shattering a set in B also permits either label on the last point. Pascal's identity completes the induction, with the empty domain as base case. For $v = 0$, all rows are identical and $K = 1$. For $1 \leq v \leq Q = |X|$, setting $t = v/Q \leq 1$ gives

$$t^v \sum_{i=0}^v \binom{Q}{i} \leq (1+t)^Q \leq e^v, \quad \ln K \leq v \ln(eQ/v) \leq (\rho/2) \ln(eQ).$$

Consequently the same learner, with no statistical-dimension conversion, has

$$\boxed{m = O(\rho^2 \ln(e|X|) + \rho \ln(1/\epsilon)), \quad \tau = \Omega(\epsilon/(\rho \ln(1/\epsilon))).}$$

For bounded ρ , this is logarithmic in the domain size at fixed accuracy. The argument is a proved consequence, not a conjectured converse characterizing SQ complexity.

4 The incidence-flip family

Let $N \geq 2$, $M = N^3$, and $K = 2M$. Points are $p = (x, y) \in [N] \times [2N^2]$. Rows are affine lines $\ell = (a, b) \in [N] \times [2N^2]$. Put $F_{\ell p} = -1$ exactly when $y = ax + b$. A monotone flip matrix A changes any subset of these -1 entries to $+1$ and leaves all other entries unchanged. In the random family the choices are independent fair bits. Denote the row class by H_A .

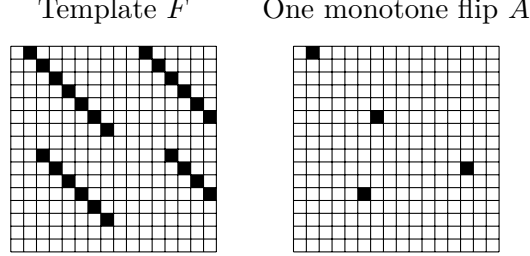


Figure 1: The $N = 2$ template and the seeded flip used in the CPU tests. Black cells are negative, white cells positive. Rows index lines; columns index points. Only black cells may change. Larger N , not this small example, yields the asymptotic counting separation.

Theorem 11 (Construction and ordinary dimension). *Every monotone flip A satisfies $\text{VC}(A) \leq 2$ and $\text{rect}(A) \geq 2^{-15}$. The template has $\text{signrank}(F) \leq 6$ and exactly*

$$I = 2N^4 - \left(\frac{N(N+1)}{2} \right)^2 \geq N^4 \quad (10)$$

independent flip positions. Set $B_0 = \log_2(4eK)$ and $s_0 = \lfloor I/(4KB_0) \rfloor$. A random A satisfies

$$\Pr[\text{dc}(H_A) \leq s_0] \leq 2^{-I/2}.$$

Every member also satisfies $\text{dc}(H_A) \leq 2N + 1$. In particular most matrices have $\text{dc}(H_A) = \Omega(N/\log M)$. All members have the learner of Theorem 9 with $\rho = 2^{15}$. Independently of the sharp rectangle input, the self-contained cap proof in Section 7 supplies $\rho = 2^{18}$ and the same asymptotic learning bound.

Proof. Define

$$u_p = (x^2, xy, y^2, x, y, 1), \quad v_\ell = (a^2, -2a, 1, 2ab, -2b, b^2 - 1/2).$$

Their dot product is $(ax + b - y)^2 - 1/2$. It is negative on incidences and positive on nonincidences because all coordinates are integers. Thus the six-dimensional strict factorization proves $\text{signrank}(F) \leq 6$. Both vectors are nonzero and may be normalized to unit vectors. Every row of A has at most N negative points. Enumerate all domain points by $t \in [K]$ and use $(1, t, \dots, t^{2N})$. For a row with negative index set S , the polynomial $\prod_{a \in S} (t - a)^2 - 1/2$ is negative exactly on S and positive at every other integer domain point. The empty product is one. This proves $\text{dc}(H_A) \leq 2N + 1$ for every member.

For fixed a, x , exactly $2N^2 - ax$ values of $b \in [2N^2]$ give an incidence with $y \in [2N^2]$. Summing proves (10). The lower bound uses $(N+1)/2 \leq N$. If two distinct affine lines share points (x, y) and (x', y') , subtraction gives $(a - c)(x - x') = 0$. If $a = c$, their shared point forces equal intercepts, a contradiction. Otherwise $x = x'$ and then $y = y'$. Hence F , and therefore A , has no all-negative 2×2 . Shattering three columns would require one row negative on all three and a distinct row negative on two of them, which creates such a submatrix. Therefore $\text{VC}(A) \leq 2$.

The homogeneous-rectangle theorem of Alon et al. [3, Theorem 1.3], in the formulation checked in Hatami et al. [2, Theorem 1.9], states that an $a \times b$ sign matrix of sign-rank at most d contains a monochromatic rectangle with each side at least the corresponding side length divided by 2^{d+1} . For rational product weights, repeat rows and columns according to common-denominator multiplicities. Projecting the resulting rectangle back to the original matrix preserves its signs and can only increase its weighted masses. For arbitrary weights, approximate by rationals; the maximum over finitely many rectangle masses is continuous. It follows that

$$\text{rect}(F) \geq 2^{-2d-2}. \quad (11)$$

Fix a product distribution and such a rectangle of F . If its color is positive, it survives every monotone flip. If its color is negative, one side is a singleton, by the absence of a negative 2×2 . Partition the other side according to its sign in A ; one part retains at least half its mass. Thus

$$\text{signrank}(F) \leq d, \quad F \text{ has no negative } 2 \times 2 \implies \text{rect}(A) \geq 2^{-(2d+3)}. \quad (12)$$

For $d = 6$ this is 2^{-15} . This reduction establishes exactly which extra structural hypothesis is needed.

For the counting bound, the theorem of Warren [4] says that $s \geq v$ real polynomials of degree at most k in v variables have at most $(4eks/v)^v$ strict sign patterns; see Alon et al. [5, Theorem 21 and Lemma 22]. A rank-at-most- d $K \times K$ matrix factors as a $K \times d$ matrix times a $d \times K$ matrix. Its K^2 entries are degree-two polynomials in $2Kd$ variables. Therefore, for $1 \leq d \leq K/2$,

$$\#\{B \in \{-1, +1\}^{K \times K} : \text{signrank}(B) \leq d\} \leq (4eK/d)^{2Kd} \leq 2^{2KdB_0}. \quad (13)$$

The random family has exactly 2^I equiprobable matrices. Taking $d = s_0$ bounds the probability by $2^{2Ks_0B_0-I} \leq 2^{-I/2}$. The range $s_0 \leq K/2$ follows from $I \leq K^2$ and $B_0 > 1$; when $s_0 = 0$ the event is empty. Finally $I \geq N^4$ and $K = 2N^3$ give the asymptotic lower bound. Apply Theorem 9 for the learner. $\square$

The restriction in (12) cannot be omitted. Every sign matrix is a monotone flip of the all-negative, rank-one matrix. More quantitatively, let W be a $k \times k$ Walsh matrix, k a power of two. Its orthogonality gives $\|W\|_{\text{op}} = \sqrt{k}$. A monochromatic rectangle of side sizes a, b satisfies

$$ab = |\mathbf{1}_S^T W \mathbf{1}_T| \leq \sqrt{k} \sqrt{ab},$$

so its uniform product mass is at most $1/k$. For $k > 32$, this contradicts the asserted unrestricted bound $2^{-(2 \cdot 1 + 3)}$.

5 Probabilistic dimension under the actual definitions

Theorem 12 (Approximate counting and exact probabilistic dimension). *Fix $0 \leq \eta < 1/2$ and write*

$$h_2(\eta) = -\eta \log_2 \eta - (1 - \eta) \log_2 (1 - \eta), \quad c_\eta = 1 - h_2(\eta) > 0, \quad s_\eta = \left\lfloor \frac{c_\eta I}{4KB_0} \right\rfloor,$$

with $0 \log 0 = 0$. For the random incidence-flip matrix,

$$\Pr[\text{dc}^\eta(H_A) < s_\eta] \leq 2^{-c_\eta I/2}. \quad (14)$$

For every finite class, independently of this construction,

$$\text{dc}(H) \leq (\lceil \log_2 K \rceil + 1) \text{dc}_{1/2}(H) + 1. \quad (15)$$

Consequently most A have $\text{dc}_{1/2}(H_A) = \Omega(N/\log^2 M)$ and $\text{dc}^\eta(H_A) = \Omega_\eta(N/\log M)$.

Proof. Let J_ℓ be the set of template incidences in row ℓ , with $k_\ell = |J_\ell|$. For each nonempty J_ℓ , take D_ℓ uniform on it. If a law on d -dimensional embeddings witnesses $\text{dc}^\eta(H_A) \leq d$, then

$$\mathbb{E}_\phi \sum_{\ell: k_\ell > 0} \frac{k_\ell}{I} \min_w L_{D_\ell, h_\ell}(\text{sign}\langle w, \phi \rangle) \leq \eta.$$

For a fixed embedding, only finitely many Boolean predictions are possible, so each infimum is attained as a minimum of finitely many loss values. There exists one embedding with the displayed weighted sum at most η . Otherwise its positive excess over η would have strictly positive expectation. Choose one minimizing weight vector for each row. Their prediction matrix makes at most ηI mistakes on the incidence positions.

With $\text{sign}(0) = +1$, this prediction matrix has strict sign-rank at most $d + 1$, not necessarily at most d . Add the constant feature one. In each row add a strictly positive bias smaller than the absolute value of every negative score in that row, if there are negative scores. This retains all signs and turns every zero into a positive score. Finiteness makes the bias choice possible. Rows with no incidence may use any weight vector.

For any fixed sign matrix B , the number of possible incidence flip patterns within ηI mistakes is at most

$$\sum_{j=0}^{\lfloor \eta I \rfloor} \binom{I}{j} \leq 2^{h_2(\eta)I}.$$

For $0 < \eta < 1/2$, this follows from

$$1 \geq \sum_{j \leq \eta I} \binom{I}{j} \eta^j (1 - \eta)^{I-j} \geq 2^{-h_2(\eta)I} \sum_{j \leq \eta I} \binom{I}{j};$$

the second inequality uses $j \leq \eta I$ and $\eta/(1 - \eta) < 1$. For $\eta = 0$ the sum is one. Combining with (13), the probability that any rank-at-most- s_η sign matrix makes at most ηI incidence errors is at most

$$2^{2Ks_\eta B_0 + h_2(\eta)I - I} \leq 2^{-c_\eta I/2}.$$

An embedding of dimension below s_η would produce just such a strict sign matrix after adding the constant coordinate. If $s_\eta = 0$, the event in question is empty. This proves (14).

For (15), choose a full-support distribution on X . Zero loss under this distribution means correct labels at every point. Draw $t = \lceil \log_2 K \rceil + 1$ independent embeddings from a law witnessing $\text{dc}_{1/2}(H) = d$. Each fixed target fails in all t embeddings with probability at most 2^{-t} . A union bound gives probability at most $K2^{-t} < 1$ that some target fails everywhere. Hence one collection works for all targets. Concatenate its t embeddings; each row selects a successful block. Adding one constant coordinate as above makes the representation strict. This proves (15). Combine with Theorem 11. If simultaneous bounds are desired, intersect its event with (14); failure probability is at most $2^{-I/2} + 2^{-c_\eta I/2}$. $\square$

For fixed $\epsilon > 0$, $\rho = 2^{15}$ gives $m = O(\log M) = O(n)$ and $1/\tau = O_\epsilon(1)$, whereas the displayed dimensions are exponential in n , up to logarithmic factors. Thus forms (a), (b), and (c) fail for dc and $\text{dc}_{1/2}$. For any proposed fixed constant $C > 0$ in $\text{dc}^{C\epsilon}$, fix $\epsilon < \min\{1/4, 1/(2C)\}$ and apply Theorem 12 at $\eta = C\epsilon$. This refutes the same three polynomial forms for that variant. The assertion $m/\tau^2 = O(\log n)$ is not used and is not implied: the proved bound is $O_\epsilon(n)$.

6 A direct check through randomized statistical dimension

We give the proposed dimension route separately. It is not needed for Theorem 9. For the search task $\mathcal{Z}_\eta$, an instance is a realizable labeled distribution $P_{D,h}$; an allowed solution is a predictor g with loss at most η . Fix any reference distribution P_0 on $Z = X \times \{-1, +1\}$, not necessarily realizable or fair-labeled. For a law $\mathcal{P}$ on predictors, put

$$\mathcal{B}(\mathcal{P}) = \{j : \Pr_{g \sim \mathcal{P}}[g \text{ is } \eta\text{-good for } j] < 1/4\}.$$

A query distribution covers this bad set at level $1/d$ when every j in it is distinguished from P_0 by more than κ with probability at least $1/d$. Taking the best $\mathcal{P}$ and query law, and the worst P_0 , is the covering formulation of Feldman's $\text{RSD}_{\kappa_1}(\mathcal{Z}_\eta, \kappa, 1/4)$.

Theorem 13 (Rectangle bound for randomized statistical dimension). *Let $L = \lceil \rho \ln(4/\eta) \rceil$ and $\kappa = \eta/(4L)$, where $0 < \eta < 1/4$. Then*

$$\text{RSD}_{\kappa_1}(\mathcal{Z}_\eta, \kappa, 1/4) \leq 4L.$$

This statement allows arbitrary target-dependent marginals and any reference P_0 .

Proof. Fix a finitely supported prior π on instances. It induces a prior ν on hypotheses. Let $\mathcal{Q}$ be the finite query family $q_{S,c}(x, y) = \mathbf{1}_S(x)\mathbf{1}_{y \neq c}$. Set

$$\beta = \max_{q \in \mathcal{Q}} \Pr_{j \sim \pi}[|P_j q - P_0 q| > \kappa].$$

Peel L rectangles using the prior-averaged example marginal restricted to the current unpeeled set U , and the unchanged hypothesis prior ν . Empty residual mass permits immediate completion. The rectangle condition gives both

$$\nu(T) \geq r, \quad \mathbb{E}_j D_j(S) \geq r \mathbb{E}_j D_j(U).$$

Suppose first $\beta < r$. On every instance with target in T , the mismatch query has expectation zero. Therefore $P_0 q \leq \kappa$: otherwise a prior mass at least r would be distinguished. Apart from a prior mass at most β , we then have $P_j q \leq 2\kappa$.

After L peels, the expected residual mass is at most $(1-r)^L \leq \eta/4$. Markov's inequality shows that at least half the prior puts residual mass at most $\eta/2$. Outside a union of exceptional sets of total mass at most $L\beta$, each of the L wrong-color masses is at most 2κ . The predictor assembled from the rectangle colors and an arbitrary residual label consequently has loss at most $2L\kappa + \eta/2 = \eta$ for prior mass at least $1/2 - L\beta$. If $\beta \geq r$, then $L\beta \geq L/\rho \geq \ln(4/\eta) > 1/2$, so the following inequality remains true trivially:

$$\max_g \Pr_{\pi}[g \text{ good}] + L \max_{q \in \mathcal{Q}} \Pr_{\pi}[q \text{ distinguishes}] \geq 1/2. \quad (16)$$

Apply the finite separation argument of Lemma 10 to the action pairs (g, q) , whose payoff on instance j is

$$\mathbf{1}_{g \text{ good for } j} + L \mathbf{1}_{q \text{ distinguishes } j}.$$

It gives laws $\mathcal{P}, \mathcal{Q}'$ satisfying

$$\Pr_{g \sim \mathcal{P}}[g \text{ good for } j] + L \Pr_{q \sim \mathcal{Q}'}[q \text{ distinguishes } j] \geq 1/2 \quad \text{for every } j.$$

On $\mathcal{B}(\mathcal{P})$, the second probability is greater than $1/(4L)$. This proves the required coverage.

The continuum of possible marginals creates no compactness gap in this minimax step. Both the predictor set and $\mathcal{Q}$ are finite. An instance is relevant only through the finite vector of Boolean good-solution and distinguishing-query indicators. There are finitely many such types. Choose one representative of each realized type and apply the finite argument to them. Its inequalities then hold for all original instances. No discretization in total variation and no factor polynomial in $|X|$ is introduced. $\square$

What the 2017 characterization actually says. In Feldman [6, Theorem 4.8], the randomized upper bound, in the notation used here, is

$$\text{RQC}(\mathcal{Z}, \text{STAT}(\kappa/3), \alpha - \delta) = O\left(d \frac{R_{\text{KL}}(\mathcal{D})}{\kappa^2} \log \frac{R_{\text{KL}}(\mathcal{D})}{\kappa\delta}\right), \quad d = \text{RSD}_{\kappa_1}(\mathcal{Z}, \kappa, \alpha). \quad (17)$$

The displayed asymptotic expression uses the usual positive-radius conventions; a ceiling and a maximum with one handle degenerate values. Here $R_{\text{KL}}(\mathcal{D}) = \inf_Q \sup_{P \in \mathcal{D}} \text{KL}(P\|Q)$. Taking uniform Q on Z proves $R_{\text{KL}} \leq \ln |Z|$: expand $\text{KL}(P\|Q) = \ln |Z| + \sum_z P(z) \ln P(z)$ and use $\ln P(z) \leq 0$. Thus the relevant domain term is $\ln(2|X|)$, not $|X|$ or the logarithm of a discretized instance count. The deterministic theorem has a different set-cover logarithm; it must not be substituted here.

For clarity, the mechanism behind (17) does not depend on a hidden finite-instance enumeration. Maintain a reference Q starting uniform on Z . A query separated from the true P by κ supplies a bounded direction with a signed expectation gap of order κ . For $f \in [-1, 1]$ and update $Q'(z) \propto Q(z)e^{\lambda f(z)}$, direct expansion gives

$$\text{KL}(P\|Q') - \text{KL}(P\|Q) = -\lambda \mathbb{E}_P f + \log \mathbb{E}_Q e^{\lambda f} \leq -\lambda(\mathbb{E}_P f - \mathbb{E}_Q f) + \lambda^2/2.$$

The final inequality follows by twice differentiating $\log \mathbb{E}_Q e^{\lambda f}$: its second derivative is a variance of a variable in $[-1, 1]$, at most one. Choosing λ proportional to κ decreases KL by $\Omega(\kappa^2)$. There are therefore $O(\ln |Z|/\kappa^2)$ updates. At each reference, sampling $O(d \ln(T/\delta))$ covering queries prevents an undetected bad instance over T updates, by $(1 - 1/d)^s \leq e^{-s/d}$ and a union bound. This proves the parameter dependence and extends the argument to all realizable marginals here.

At fairness $1/4$ and constant $\delta < 1/4$, this yields constant success probability. Fresh repetitions and a single loss query to verify each candidate amplify success with $O(\ln(1/\delta'))$ repetitions and loss slack $O(\kappa)$. This reasoning remains valid for history-dependent oracles by conditioning on earlier runs. The verifiable-search Theorem 4.11 in Feldman [6] is stated using a different statistical dimension over references admitting no good solution; its dimension must not be identified automatically with the fairness- $1/4$ quantity above. In either route verification does not erase the R_{KL}/κ^2 term of that generic simulation. The direct learner of Theorem 9 gives the sharper bound without it.

7 The geometric proposal: repairs and attacks

A cap must be defined by its angular radius, not by an ambiguous “height.” With $\alpha = (2d)^{-1/2}$ and unit u , use

$$C_u = \{z \in S^{d-1} : \langle u, z \rangle \geq \sqrt{1 - \alpha^2}\}.$$

For unit v with $\langle u, v \rangle > \alpha$, every $z \in C_u$ has $\langle z, v \rangle > 0$. Indeed, writing $t = \langle u, z \rangle$ and $s = \langle u, v \rangle$, Cauchy–Schwarz gives

$$\langle z, v \rangle \geq ts - \sqrt{1 - t^2} \sqrt{1 - s^2} > 0,$$

because $t, s > 0$ and $t^2 + s^2 > 1$. The analogous negative statement follows by replacing v with $-v$.

If ν is in normalized radial-isotropic position, then

$$\mathbb{E}_\nu \langle u, v \rangle^2 = 1/d, \quad \nu\{|\langle u, v \rangle| > \alpha\} \geq 1/d - \alpha^2 = 1/(2d).$$

The second inequality follows by upper-bounding the square by α^2 on the complement and by one elsewhere. The actual isotropy condition uses $Tv/\|Tv\|$; ordinary covariance whitening followed by normalization does not imply it. When a cap contains at least two template points, at most one line belongs to the negative side, by the intersection property. If every row has prior weight below $1/(4d)$, the positive side has mass at least $1/(4d)$ and avoids every point of that cap.

These local facts do not justify discarding hypotheses outside a heavy subspace. Such a restriction can discard the true row and destroys the potential argument. For the uniform surviving priors used by the learner, one repair is to perturb the finite strict representation once into general position, preserving all template signs. If more than d rows survive, every d surviving row vectors are independent, and the radial-isotropy argument in the proof of retained Theorem 3 applies. When at most $4d$ rows survive, direct loss tests suffice, with a separately allocated error budget. This is a valid way to remove subspace degeneracy from a cap-based proof; it does not make the unmodified subspace restriction safe.

The proposed candidate-test threshold also needs tightening. An answer at least $1 - 2\epsilon - 2\tau$ certifies only correlation at least $1 - 2\epsilon - 3\tau$, hence error at most $\epsilon + 3\tau/2$. The proposed stop tolerance also permits residual mass $5\epsilon/8$, while its peel allocation permits total wrong mass $\epsilon/2$; their sum is $9\epsilon/8$, so those constants do not prove the claimed final error. Constant expected retries do not constitute a worst-case query bound. A singleton label must be queried as a bounded weighted query, not as a free conditional-label oracle. The rectangle learner avoids these issues and proves a global round-tail bound.

Proposition 14 (Independent constant rectangle bound). *Let c_d be the normalized surface measure of the preceding cap on S^{d-1} , $d \geq 2$. Every matrix F of sign-rank at most d satisfies $\text{rect}(F) \geq c_d/(4d)$. If F has no negative 2×2 , every monotone flip A satisfies $\text{rect}(A) \geq c_d/(8d)$. In particular, for $d = 6$ this is greater than 2^{-18} .*

Proof. First take a uniform prior on $k > d$ row copies. Perturb their representing vectors, if necessary, so every d are independent while preserving the finitely many strict signs. The full minimization argument in retained Theorem 3 produces an invertible transform whose normalized row vectors are isotropic. Apply the inverse transpose to the point vectors and normalize; all signs persist.

For each uniform random direction u , the two sets $T_+(u) = \{v : \langle u, v \rangle > \alpha\}$ and $T_-(u) = \{v : \langle u, v \rangle < -\alpha\}$ have combined prior mass at least $1/(2d)$. The rectangles $C_u \times T_+(u)$ and $C_u \times T_-(u)$ have constant signs. For any fixed point marginal μ , rotational invariance gives $\mathbb{E}_u \mu(C_u) = c_d$, because each unit point lies in a random cap with probability c_d . Therefore the expected sum of their product masses is at least $c_d/(2d)$, and some single rectangle has mass at least $c_d/(4d)$.

For a rational row prior, duplicate rows with common-denominator multiplicities, repeating all copies further to ensure $k > d$. The transformed representation may differ across identical copies, but every pairwise sign still equals the original matrix entry. Projecting a rectangle to original rows only increases its prior mass. Approximation by rational priors and continuity of the finite maximum prove the assertion for all priors. A negative rectangle has a singleton side under the additional assumption on F , so the same half-mass argument as in (12) proves the second assertion.

For the numerical estimate at $d = 6$, project the cap onto its last five coordinates. Its image is the five-dimensional ball of radius $\alpha = 1/\sqrt{12}$, and the surface-area Jacobian is at least one. The

volume of the unit five-ball is $8\pi^2/15$ and the area of S^5 is π^3 , obtained by polar integration of the Gaussian integral. Thus

$$c_6 \geq \frac{8}{15\pi 12^{5/2}} > \frac{8}{15 \cdot 4 \cdot 576} = \frac{1}{4320}, \quad \frac{c_6}{48} > \frac{1}{207360} > 2^{-18}.$$

This independent constant, rather than the sharper imported one, is already sufficient for Theorems 9–12 to refute every polynomial dimension implication. $\square$

Where the proposed attacks are absorbed. Under a marginal uniform on the target line, many geometric positive rectangles miss most of the mass. They may be rejected, but the pointwise mixture inequality forces conditional expected progress at least $r/4$; no minimum geometric margin under D is assumed. Under a marginal supported only on the target’s negative points, a positive rectangle with substantial mass has nonzero wrong-color mass and therefore cannot eliminate the target. It either safely eliminates other rows or is peeled only within the 3τ error allowance. When mass is split between negative and flipped points of the same template line, it is the labels of A , not of F , that define the learner’s monochromatic rectangles and mismatch query. Every flipped label is accounted for.

For two- or three-point support, all queries remain unconditioned indicators. Small point masses are never divided out of a label query. A low-mass tail is paid for in the residual error budget. A prior supported on a parallel class causes no singularity in the rectangle LP and removes no target without a mismatch witness. Finally, a distribution chosen adversarially against a geometric cap sampler still satisfies the pointwise coverage inequality. It may change the actions and number of rejections, but cannot change the drift calculation. These are uniform oracle-level guarantees, not conclusions from the finite experiments.

8 Why spectral certificates could not see this

Proposition 15 (Uniform-product rectangles in a projective plane). *For the point–line incidence sign matrix of $\text{PG}(2, q)$, let $K = q^2 + q + 1$ and $k = q + 1$. Under uniform distributions on both sides, the largest monochromatic-rectangle mass lies between*

$$\frac{1}{8(q+1)} \quad \text{and} \quad \frac{q}{(q+1)^2}.$$

It is therefore $\Theta(1/q)$.

Proof. Let P be the incidence matrix. As proved in retained Theorem 4, $P\mathbf{1} = k\mathbf{1}$ and $P^T P = qI + \mathbf{1}\mathbf{1}^T$. Thus $C = P - (k/K)\mathbf{1}\mathbf{1}^T$ has operator norm $\sqrt{q}$. A nonincident rectangle with side sizes a, b satisfies

$$\frac{k}{K}ab = |\mathbf{1}_S^T C \mathbf{1}_T| \leq \sqrt{q}\sqrt{ab},$$

so its mass is at most q/k^2 . An all-incident rectangle has a singleton side and size at most k . Its mass k/K^2 is also at most q/k^2 : use $K \geq qk$ and $q^3 \geq k$ for $q \geq 2$ to obtain $qK^2 \geq q^3k^2 \geq k^3$.

For the lower bound, choose $s = \lfloor K/(2k) \rfloor$ lines. Their union has at most $sk \leq K/2$ points, leaving at least $K/2$ points incident with none of them. The resulting rectangle has mass at least $s/(2K)$. Since $K/(2k) \geq 1$, $s \geq K/(4k)$, proving the lower bound. $\square$

The global rectangle ratio is an infimum, so this computation gives $\text{rect}(\text{PG}(2, q)) \leq O(1/q)$; it does not by itself prove the same lower bound for every product distribution. The joint certificate in retained Theorem 4 has strength $\Omega(q)$ and therefore remains consistent with the rectangle learner.

For the incidence-flip family, apply retained Theorem 2 to the learner at any fixed accuracy. Every joint-distribution Gram ratio J in that theorem satisfies

$$J \leq 4 + 4m/\tau^2 = O(\log K).$$

At the same time most matrices have dimension $\Omega(K^{1/3}/\log K)$. Hence the earlier proposed completeness inequality $\text{dc}(H) \leq \text{poly}(\mathfrak{J}(H), \log |X|)$ is false, not merely unsupported. The rectangle method proves an upper bound on learning complexity precisely where the sign-rank lower bound comes from counting rather than an ordinary spectral certificate. We make no blanket claim equating every possible spectral method with a single rectangle parameter.

For comparison with Hatami et al. [2, Proposition 3.14], every matrix with $\text{rect} \geq r$ has deterministic communication of expected cost at most $2/r$ under each fixed product distribution. At a node, choose a rectangle of conditional product mass at least r . The two parties send the two membership bits. On success output its color; on failure condition on the resulting nonempty subrectangle and repeat. Conditional distributions remain products. The chance of continuing for at least t stages is at most $(1 - r)^t$, so expected stages are at most $\sum_{t \geq 0} (1 - r)^t = 1/r$. This is a different protocol for each product distribution. Lemma 10 is the step that removes knowledge of the example marginal from the learning algorithm.

9 The SGD question and the exact transfer

To place the class on a Boolean cube, let $b = \lceil \log_2 K \rceil$, identify the K original points with a subset of $\{-1, +1\}^b$, and give every row label $+1$ on the unused points. If a marginal gives the unused points mass at least $1/2$, they form a positive rectangle of product mass at least $1/2$. Otherwise the original points carry more than half the mass, and their conditional rectangle guarantee gives a rectangle of mass at least $r/2$. Hence the padded class has $\text{rect} \geq r/2$. Restricting embeddings to the original points shows that each of its three dimension notions is at least the corresponding original one. The SQ separation therefore also holds on a Boolean cube.

For an adaptively chosen scalar query q , the objective $J_q(t; x, y) = tq(x, y)$ has derivative $q(x, y)$ at every t . Its population derivative is $\mathbb{E}q$. Thus an approximate scalar-gradient oracle is exactly an SQ oracle. This proves a transfer for adaptively selected objectives, not for one fixed conventional network.

The converse coordinate simulation remains as in the earlier draft. A robust population-gradient method with S coordinates bounded per example by B_g , T calls, and adversarial Euclidean gradient tolerance $\delta > 0$, is simulated by TS SQ queries of tolerance $\delta/(B_g\sqrt{S})$. Divide each coordinate by B_g and query it while keeping the state fixed; summing the S squared coordinate errors gives at most δ^2 . Induction preserves the robust guarantee. This is an exact proved statement, but it cannot now be combined with a universal polynomial SQ-to-dimension theorem, which the preceding construction refutes.

Abbe et al. [8, Theorem 1d and Sections 4–5] give SQ-to-mini-batch differentiable-learning simulations using deliberately constructed models and initialization. Their computational neural-network bound depends on the running time of the simulated algorithm, not just its number of SQ calls. Their piecewise-linear activation has a flat central region; their ReLU alternative uses weights held fixed during training. The construction here therefore transfers to some engineered differentiable model, and with its finite computational implementation to some engineered network.

It does not produce a small network merely because $m = O(n)$. Both architecture/initialization restrictions and computational representation size remain substantive hypotheses.

The following question is unchanged, rather than answered by the simulation. Let $X = \{-1, +1\}^n$. Consider a fully connected bias-free ReLU network of depth L , widths $n_0 = n, n_1, \dots, n_L = 1$, and S trainable weights. Initialize every layer- i weight independently as $N(0, 1/n_{i-1})$. At each of T online steps, draw a fresh $x \sim D$ and update all weights by gradient descent with step size η on $\log(1 + e^{-h(x)f_\theta(x)})$. Output the sign of the sum of the networks in the final half of the trajectory, fixing a ReLU derivative convention at zero.

Open problem (Benign online SGD versus dimension). *If this one architecture, step size, and step count achieve expected error below $1/4$ for every D and every target in H , must $\text{dc}(H) \leq CTS$, or at least $\text{poly}(T, S)$? No programmable initialization, fixed-weight circuit, alternative objective, or adversarial population-gradient guarantee is inserted into the premise.*

To refute this for the present family requires a small such architecture together with a proof about the specified random initialization and online dynamics for all marginals. To prove it requires a consequence of these dynamics not possessed by general distribution-adaptive SQ algorithms. A sampled exact gradient need not be close to the population gradient, and success in expectation over sampled gradients does not imply robustness to every bounded perturbation. Gaussian initialization alone supplies no uniform gradient-coordinate bound. No online-SGD experiment is included in this turn.

10 Explicitness and current topological results

The random-flip selection is not an explicit deterministic sequence of high-sign-rank truth tables. The learner, given any table in the family, is explicit. These are separate meanings of explicitness.

The Borsuk–Ulam lower bounds of Hatami et al. [9] concern partial gap-inner-product and Gap Hamming matrices. The 2026 framework of Frick et al. [11] proves

$$\text{signrank}(\text{GHD}_k^b) = (1 - o_k(1)) 2k, \quad o_k(1) = O\left(\sqrt{\log k/k}\right).$$

Here $k < b/2$ and the matrix is partial. This lower bound is at most linear in the bit length $b = \log_2 |X|$, not exponential in a positive power of b . A completion using the sign of the Hamming inner product has dimension at most $b + 1$: represent the b sign coordinates and a constant bias, choosing a bias to avoid zero while preserving the promised near/far signs. Thus these matrices do not provide the requested explicit total counterexample by themselves.

Moreover, Göös et al. [10] prove that the total k -Hamming-distance matrices have sign-rank $2^{O(k)}$ independent of the number of bits. In particular, the fixed- k hypercube candidate suggested in the older literature does not have growing dimension. The checked 2023–2026 sources do not supply an explicit total matrix with bounded inverse rectangle ratio and dimension $2^{\Omega(n^c)}$. We leave that explicit strengthening open, in the direction of Problem 4.1 of Hatami et al. [2]. This is a statement about what these checked results provide, not a claim to have exhaustively classified all recent papers.

11 Executed CPU verification

The script `experiments/rectangle_sq.py` constructs F , samples independent incidence flips, checks the six-coordinate factorization integrally, and implements the rectangle learner with exact

rational oracle answers. It uses a finite rectangle catalogue. SciPy proposes a mixture, which is rounded to integer weights. Before any sampling, integer arithmetic verifies

$$\min_{x \in U} \sum_j w_j |T_j \cap V| \mathbf{1}_{S_j}(x) \geq \frac{|V| \sum_j w_j}{32768}.$$

Thus a floating-point LP optimum is not used as an unverified feasibility assumption. The catalogue always includes both colored singleton-point rectangles; for $|X| \leq 250$ these alone certify coverage at least $1/(2|X|) > 1/32768$. Additional rectangles improve practical progress. Every stored mixture certificate can be independently rechecked from its rectangle supports and integer weights. The full theoretical learner enumerates all rectangles; the tested subcatalogue is valid precisely because its coverage was certified.

This is a replacement of the requested isotropic-transform experiment, not an implementation of an exact isotropic transform. It tests the learner actually proved here. The tolerance is exactly $1/137846880$, with $\epsilon = 1/10$, $\rho = 32768$, and $P = 574362$. All oracle answers are fractions of integral distribution weights. The seeded search uses supports of size at most 20, separate objectives for error and round count, four random restarts per objective, ten coordinate steps per restart, and two candidate mutations per step. Fixed attack distributions and five disjoint held-out learner seeds are also included. The search is empirical, not a certificate of worst-case error.

N	$ X $	Runs	Max. loss	Max. rounds	Peel %	Eliminate %	Reject %
2	16	199	0.024999	8	62.21	33.16	4.63
3	54	198	0.024999	15	35.51	46.54	17.95
4	128	198	0.024999	22	34.31	32.63	33.05
5	250	198	0.024999	30	34.60	36.13	29.27

An independent audit script rechecks all 393 stored mixture laws at 29,242 point constraints using integers, all 793 trajectories and 3,863 proposal rounds using exact fractions, and all 26 small sign-rank records. It also checks 20,000 endpoint cases of the adversarial-tolerance acceptance inequalities, 1,000 finite peeling partitions, 1,000 elimination sequences, 800 entropy-ball inequalities, and 500 spectral energy cases. The transcendental and spectral numerical checks use explicit floating tolerances; they are not claimed as proofs of their infinite-instance counterparts. Every assertion passed. The largest observed loss at each N is exactly $24999/10^6$, coming from a deliberately constructed small-negative-mass attack. This is an observed maximum, not a certified worst case.

A proposal round is classified as a peel, elimination, or rejection. Candidate-verification tests occupy zero rounds because the replacement algorithm does not use them. Stopping mass queries are counted in total queries, but not as proposal rounds. Raw records contain the support, integer weights, target, seed, exact loss, every action, and each mixture-certificate identifier.

For $N = 2$, every row's negative set has at most two points. The graph whose edges are the two-point negative sets of F is a disjoint union of paths. Each edge joins $(1, y_1)$ to $(2, y_2)$ with $y_2 - y_1 \in \{1, 2\}$, so degrees are at most two. Along any nonbacktracking walk, the successive y_1 coordinates move strictly in one direction by one at a time; reversing direction would require using the preceding edge. Thus no cycle exists. Ordering vertices along these paths makes each negative pair consecutive. If a point has position t , use $(1, t, t^2)$. A negative singleton at position i is realized by $2(t - i)^2 - 1$; a negative pair at $i, i + 1$ is realized by $4(t - i - 1/2)^2 - 4$; an empty negative set is realized by one. These polynomials have the claimed signs at every integer position, so they are exact rank-three upper certificates for every monotone flip at $N = 2$.

There is an all-positive row. In any strict rank-two representation it supplies an affine chart on the point vectors, after positive scaling. Thus the negative set of each other row is a prefix or

suffix of one ordering. Conversely, such an ordering is realized by affine thresholds and gives rank at most two. The program exhausts, for each distinct proper negative set, the choice of itself or its complement as a prefix, and checks whether all selected sets form a nested chain. Complementing every choice is redundant, so one choice is fixed. Every feasible chain supplies an ordering by successive set differences. This is an exact exhaustive rank-two test, not a numerical factorization heuristic. Rank one is possible exactly when all rows are constant, because of the all-positive row. The 24 sampled flip matrices and F all have exact sign-rank three; the all-flipped control has sign-rank one. All certificates and enumeration counts are saved.

12 Finite Lean targets and verification status

The definitions of finite distributions, strict dimension, SQ queries, adaptive real-response decision trees, oracle validity, and distribution-independent randomized learning remain in `SQDC.lean`. Four additional finite statements are in `Turn2.lean`. The Lean sources encode labels as `Bool`, using `true` for $+1$ and `false` for -1 . They are stated here to make the source-to-paper correspondence exact.

Lemma L1 (Peeling error). *Let $o : X \rightarrow \{\perp\} \cup [t]$ assign every point to at most one peel. Let $c_i \in \{-1, +1\}$ and let g_0 be any fallback. Define $g(x) = c_i$ if $o(x) = i$, and $g(x) = g_0(x)$ otherwise. Put $w_i = D\{x : o(x) = i, c_i \neq h(x)\}$ and $u = D\{x : o(x) = \perp\}$. Then $L_{D,h}(g) \leq \sum_i w_i + u$. If $w_i \leq 3\tau$ and $u \leq u_0$, then $L_{D,h}(g) \leq 3t\tau + u_0$.*

Proof. Partition the finite sum defining loss by $o(x)$. On a peeled set its contribution is exactly w_i ; on the residual set each error indicator is at most one. Sum the pointwise inequalities and then the upper bounds on w_i . $\square$

Lemma L2 (Elimination potential). *Let $p_0, \dots, p_R > 0$, $p_R \leq 1$, and $0 \leq b_t < 1$ with $p_{t+1} = p_t/(1 - b_t)$ for $0 \leq t < R$. Then $\sum_{t < R} b_t \leq -\ln p_0$.*

Proof. The inequality $\ln(1 - b_t) \leq -b_t$ gives $b_t \leq \ln p_{t+1} - \ln p_t$. Sum and use $\ln p_R \leq 0$. $\square$

Lemma L3 (Positive rectangles survive). *If $F_{ij} = +1$ implies $A_{ij} = +1$ for every entry, every positive rectangle of F is a positive rectangle of A .*

Proof. Apply the entrywise implication at every pair in the rectangle. $\square$

Lemma L4 (Two integer grid lines). *Let $a, b, c, d, x, y, x', y' \in \mathbb{Z}$ with $(a, b) \neq (c, d)$, $y = ax + b = cx + d$, and $y' = ax' + b = cx' + d$. Then $x = x'$ and $y = y'$.*

Proof. If $a = c$, the first equality gives $b = d$, a contradiction. Subtracting the two shared-point equalities gives $(a - c)(x - x') = 0$. The integers have no zero divisors, so $x = x'$, and substitution gives $y = y'$. $\square$

The Lean scripts are unverified. The actual commands `lake -version`, `lake build`, and `lean -version` each returned command-not-found and exit 127. The source is pinned to Mathlib commit 2631d1cc8c2ace6c6a900425d6e5d2b5963966e9, whose toolchain file specifies `leanprover/lean4:v4.34.0-rc2`. The repository commit and toolchain were read through the GitHub connector. The scripts contain no deliberately inserted `sorry` or custom axioms, but no successful compilation or axiom audit is claimed. The main randomized learner and the real-algebraic counting theorem are not yet formalized. Build instructions and actual failure logs are included.

13 What remains open

The SQ-to-dimension implications in (a), (b), and (c), including the two probabilistic variants at fixed nontrivial accuracy, are refuted by the preceding arguments. The remaining mathematical questions are an explicit total counterexample, a computationally efficient version with small query circuits, the benign online-SGD question, and a converse characterizing distribution-independent SQ complexity by a suitable rectangle-type measure. Theorem 9 proves sufficiency of the product-rectangle condition, not necessity. The unconditional compiled Lean formalization is a remaining deliverable, not an established result.

A Retained barrier, plane, and transcript results

The following statements and proofs retain the numbering of the first draft. They remain compatible with the counterexample: the barrier is a lower bound from a certificate, not a completeness theorem for all large dimension. In this appendix $m \geq 1$, $0 < \tau \leq 1$, and $\epsilon < 1/4$ when those restrictions are used.

Lemma 1 (Finite energy and threshold counting). *Let P_0 be a probability distribution on a finite set Z , and let $r_1, \dots, r_K \in L^2(P_0)$. Set*

$$\Gamma_{ij} = \mathbb{E}_{P_0} r_i r_j, \quad \Lambda = \|\Gamma\|_{\text{op}}.$$

For every $a : Z \rightarrow \mathbb{R}$,

$$\sum_{j=1}^K (\mathbb{E}_{P_0} r_j a)^2 \leq \Lambda \mathbb{E}_{P_0} a^2. \quad (18)$$

In particular, if $|a| \leq 1$ and $\tau > 0$, then

$$|\{j : |\mathbb{E}_{P_0} r_j a| > \tau\}| \tau^2 \leq \Lambda. \quad (19)$$

More generally, for any finite list of real numbers c_i , $\tau \geq 0$, and $\sum_i c_i^2 \leq \Lambda$, one has $|\{i : \tau < |c_i|\}| \tau^2 \leq \Lambda$.

Proof. Let $T : \mathbb{R}^K \rightarrow L^2(P_0)$ send b to $\sum_j b_j r_j$. Then $T^*T = \Gamma$. For every unit vector b ,

$$|\langle b, T^*a \rangle| = |\langle Tb, a \rangle| \leq \|Tb\|_2 \|a\|_2 \leq \sqrt{\Lambda} \|a\|_2.$$

Taking the supremum over b and squaring proves (18). Each selected summand in (19) is greater than τ^2 ; discarding nonnegative unselected summands proves the result. The same argument, with weak inequalities, proves the final statement even when $\tau = 0$. $\square$

Theorem 2 (Joint-distribution SQ barrier). *Let μ be a full-support distribution on X and let $P_0 = \mu \otimes \text{Unif}\{-1, +1\}$. For $j \in [K]$, let P_j be the labeled distribution arising from some D_j and $h_j \in \mathcal{H}$. Put*

$$r_j = \frac{P_j}{P_0} - 1, \quad \Gamma_{ij} = \mathbb{E}_{P_0} r_i r_j, \quad \Lambda = \|\Gamma\|_{\text{op}}, \quad R = K/\Lambda.$$

Any randomized adaptive (m, τ) -SQ learner satisfying (1) obeys

$$1 - 2\epsilon \leq \sqrt{\frac{\Lambda}{K}} + \frac{m\Lambda}{K\tau^2}, \quad \text{equivalently} \quad \frac{m}{\tau^2} \geq (1 - 2\epsilon)R - \sqrt{R}. \quad (20)$$

Consequently, for $\epsilon < 1/4$,

$$R \leq 4 + 4m/\tau^2. \quad (21)$$

Proof. First $\Lambda > 0$. Indeed, with $a_j(x, y) = yh_j(x)$ we have $\|a_j\|_{L^2(P_0)} = 1$ and $\langle r_j, a_j \rangle = \mathbb{E}_{P_j} a_j - \mathbb{E}_{P_0} a_j = 1$. Thus $\|r_j\|_2 \geq 1$ and $\Lambda \geq \Gamma_{jj} \geq 1$.

For a fixed private seed, run the learner on the reference responses $\mathbb{E}_{P_0} q$. Write the resulting queries as $q_1, \dots, q_m$ and the leaf output as g_0 . This is a counterfactual tree path; the reference need not be realizable. For target j , choose the following memoryless oracle, which does not know the private seed:

$$O_j(q) = \begin{cases} \mathbb{E}_{P_0} q, & |\mathbb{E}_{P_j} q - \mathbb{E}_{P_0} q| \leq \tau, \\ \mathbb{E}_{P_j} q, & \text{otherwise.} \end{cases}$$

It is valid for P_j on every query. If its run first departs from the reference transcript, that departure occurs at one of the reference queries with $|\langle r_j, q_t \rangle| > \tau$. Lemma 1 and a union bound give, for this fixed seed,

$$\frac{1}{K} |\{j : \text{the transcript departs from the reference}\}| \leq \frac{m\Lambda}{K\tau^2}.$$

For the reference leaf, define $c_j = \mathbb{E}_{P_j}[yg_0(x)]$. Since $\mathbb{E}_{P_0}[yg_0(x)] = 0$, Lemma 1 gives

$$\frac{1}{K} \sum_j |c_j| \leq \sqrt{\frac{1}{K} \sum_j c_j^2} \leq \sqrt{\Lambda/K}.$$

Let s_j be the correlation of the actual output under O_j . On a nondeparting transcript it equals c_j ; on a departing transcript it is at most 1. Therefore $s_j \leq |c_j| + \mathbf{1}_{\text{departure}}$ for j . Average this inequality over j and then over the private seed. The learner's guarantee gives an average correlation at least $1 - 2\epsilon$. This proves (20). Notice that we did not choose a different oracle after seeing the seed.

Finally write $M = m/\tau^2$. From (20), $R/2 \leq \sqrt{R} + M$. The square $(\sqrt{R} - 2)^2 \geq 0$ gives $\sqrt{R} \leq R/4 + 1$. Combining yields (21). $\square$

The requested fixed-marginal barrier. Set $D_j = \mu$ for every j . Then $r_j(x, y) = yh_j(x)$ and

$$\Gamma_{ij} = G_{ij} = \mathbb{E}_\mu[h_i h_j].$$

For $R = K/\|G\|_{\text{op}} \geq 16$, (20) implies

$$m \geq \frac{1}{4} R \tau^2. \quad (22)$$

Indeed, $1 - 2\epsilon > 1/2$ and $\sqrt{R} \leq R/4$. The cutoff matters: a uniformly positive lower bound proportional to $R\tau^2$ cannot hold for all tiny families, since a singleton needs no queries. The next theorem supplies the accompanying dimension lower bound.

A.1 The retained sign-rank lower bound

We use the spectral method associated with Forster [7], but include its geometric step rather than treating the published theorem as a proof.

Theorem 3 (Spectral sign-rank bounds). *If $A \in \mathbb{R}^{N \times K}$ satisfies $|A_{xj}| \geq 1$ for all entries, then*

$$\text{signrank}(\text{sign } A) \geq \frac{\sqrt{NK}}{\|A\|_{\text{op}}}. \quad (23)$$

If $M \in \{-1, +1\}^{N \times K}$ and $G = M^\top \text{diag}(\mu)M$ for any distribution μ on its rows, then

$$\text{signrank}(M) \geq \sqrt{K/\|G\|_{\text{op}}}. \quad (24)$$

Proof. Let $d = \text{signrank}(M)$, with $M = \text{sign } A$ in the first assertion. Choose a strict representation $M_{xj} = \text{sign}\langle z_x, v_j \rangle$ in $\mathbb{R}^d$. If $d \geq K$, both conclusions are immediate: $\|A\|_{\text{op}} \geq \sqrt{N}$ from any column, while $G_{jj} = 1$ implies $\|G\|_{\text{op}} \geq 1$. We henceforth have $1 \leq d < K$.

Perturb the finitely many v_j within sufficiently small open neighborhoods so that signs are unchanged and every d of them are independent. Such a perturbation exists: for each d -subset the determinant is a nonzero polynomial in the vector coordinates, and a finite union of its zero sets has empty interior. To justify this elementary fact, a nonzero univariate polynomial has finitely many roots; induction on the number of variables, by viewing the polynomial as a polynomial in the last variable, shows that a multivariate polynomial cannot vanish on a nonempty open box. Apply this to the product of the finitely many determinant polynomials.

We prove that there is an invertible T such that the normalized vectors $u_j = Tv_j/\|Tv_j\|$ satisfy

$$\sum_{j=1}^K u_j u_j^\top = (K/d)I_d. \quad (25)$$

For every $1 \leq r \leq d$, there is a common $\delta > 0$ such that at most $d - r$ of the v_j have projection norm less than δ onto any r -dimensional subspace. In fact, any $d - r + 1$ of the vectors are independent, so cannot all belong to the orthogonal complement of an r -dimensional subspace. Their maximum projection norm is therefore positive for every such subspace. Its minimum is positive by compactness of the space of orthogonal projection matrices of rank r ; this space is closed and bounded. Take the minimum over finitely many subsets and r .

For positive definite Q with $\det Q = 1$, minimize

$$F(Q) = \sum_{j=1}^K \log(v_j^\top Q v_j).$$

Write the decreasing eigenvalues of Q as $e^{s_1} \geq \dots \geq e^{s_d}$, where $\sum_i s_i = 0$. For each r , at least $K - d + r$ of the quadratic forms are at least $\delta^2 e^{s_r}$. Order the K log quadratic forms increasingly. For the first d entries use respectively the bounds $\log \delta^2 + s_d, \dots, \log \delta^2 + s_1$; each of the remaining $K - d$ entries is at least $\log \delta^2 + s_1$. Thus

$$F(Q) \geq K \log \delta^2 + (K - d)s_1.$$

Since $K > d$, every sublevel set has bounded largest eigenvalue. The determinant condition then bounds the smallest eigenvalue away from zero. Sublevel sets are compact, so a minimizer Q exists. For a symmetric traceless matrix B , the path $Q(t) = Q^{1/2} \exp(tB) Q^{1/2}$ retains determinant one. Its derivative at zero gives

$$0 = \sum_j \frac{v_j^\top Q^{1/2} B Q^{1/2} v_j}{v_j^\top Q v_j} = \text{tr} \left(B \sum_j u_j u_j^\top \right), \quad u_j = \frac{Q^{1/2} v_j}{\|Q^{1/2} v_j\|}.$$

A symmetric matrix orthogonal to every traceless symmetric matrix is scalar: subtract its scalar trace part and take that difference as B . Taking traces gives (25), with $T = Q^{1/2}$.

Replace each row vector by the normalization of $T^{-\top} z_x$, denoted e_x , so $\|e_x\| = 1$ and all signs are preserved. Let U be the matrix with columns u_j . For every row of A ,

$$\langle e_x, U A_{x,\cdot}^\top \rangle = \sum_j |A_{xj}| |\langle e_x, u_j \rangle| \geq \sum_j \langle e_x, u_j \rangle^2 = K/d.$$

Here $|\langle e_x, u_j \rangle| \leq 1$ was used. Consequently,

$$NK^2/d^2 \leq \|UA^\top\|_F^2 \leq \|A\|_{\text{op}}^2 \|U\|_F^2 = K\|A\|_{\text{op}}^2.$$

This proves (23). For M , use the same row inequality, square, and average under μ :

$$K^2/d^2 \leq \sum_x \mu(x) \|UM_{x,\cdot}^\top\|^2 = \text{tr}(UGU^\top) \leq \|G\|_{\text{op}} \text{tr}(UU^\top) = K\|G\|_{\text{op}}.$$

The last inequality follows by diagonalizing the positive semidefinite G , whose eigenvalues lie between zero and its operator norm. This proves (24). $\square$

Theorems 2 and 3 prove the valid conditional assertion in the proposed spectral barrier: a certificate with ratio R yields dimension lower bound $\sqrt{R}$ and query lower bound $\Omega(R\tau^2)$ when R is large. They do not prove that all explicit sign-rank lower bounds admit such an ordinary label-Gram certificate. The following family disproves that extra assertion in precisely this form.

A.2 Projective planes

Theorem 4 (An escaping plain-Gram family, but not an SQ separation). *For each prime $q \geq 2$, let X_q be the points of the projective plane over $\mathbb{F}_q$, and let $\mathcal{H}_q$ have one hypothesis per line, positive on that line and negative elsewhere. Set $N = q^2 + q + 1$ and $k = q + 1$. Then:*

$$\frac{q+1}{\sqrt{q}} \leq \text{dc}(\mathcal{H}_q) \leq 2q+3, \quad (26)$$

$$\frac{|J|}{\|[\mathbb{E}_\mu h_i h_j]_{i,j \in J}\|_{\text{op}}} \leq 16 \quad \text{for every distribution } \mu \text{ and nonempty } J \subseteq \mathcal{H}_q. \quad (27)$$

There is a distribution-independent N -query tolerance- τ learner of loss at most τ . For each fixed D and $0 < \epsilon < 1/4$, there is a learner allowed to depend on D , with tolerance $\epsilon/2$ and

$$m_D \leq 1 + s + \binom{s}{2}, \quad s = \left\lceil \frac{4 \log N + 2}{\epsilon} \right\rceil,$$

that has loss at most ϵ for every target. Here $\log$ is natural. Nevertheless every single learner satisfying (1), with $m \geq 1$, $0 < \tau \leq 1$, and $\epsilon < 1/4$, satisfies

$$\text{dc}(\mathcal{H}_q) \leq 19m/\tau^2. \quad (28)$$

Proof. Points are one-dimensional subspaces of $\mathbb{F}_q^3$; lines are kernels of nonzero linear functionals, modulo nonzero scalar multiples. There are $(q^3 - 1)/(q - 1) = N$ of each. A line contains $(q^2 - 1)/(q - 1) = k$ points. Two distinct points span a unique two-dimensional subspace, hence lie on one common line. Two distinct kernels intersect in one one-dimensional subspace. These facts also show each point lies on k lines.

Let P be the $N \times N$ incidence matrix. Then

$$P\mathbf{1} = k\mathbf{1}, \quad P^\top P = qI + \mathbf{1}\mathbf{1}^\top.$$

Define $A = (N/k)P - \mathbf{1}\mathbf{1}^\top$. Its positive entries equal $N/k - 1 = q^2/(q+1) \geq 1$, and its negative entries equal -1 . Direct multiplication gives

$$A\mathbf{1} = 0, \quad A^\top A = (N/k)^2 q \left(I - \frac{\mathbf{1}\mathbf{1}^\top}{N} \right).$$

Indeed, the coefficient of $\mathbf{1}\mathbf{1}^\top$ before simplification is $N^2/k^2 - N = -Nq/k^2$, since $k^2 = N + q$. Thus $\|A\|_{\text{op}} = (N/k)\sqrt{q}$, and Theorem 3 gives the lower bound in (26).

For the upper bound, enumerate the points by distinct integers $1, \dots, N$ and use the common embedding $\phi(t) = (1, t, \dots, t^{2k})$. For a positive set S of size k , the polynomial

$$p_S(t) = \frac{1}{2} - \prod_{a \in S} (t - a)^2$$

is positive on S and negative on all other domain points: away from S the product is a positive integer. Its coefficient vector is the required weight vector. Hence $\text{dc} \leq 2k + 1$.

Now prove (27). Write $K = |J|$, and restrict P to the selected columns. If $K < 16$, the Gram matrix has diagonal entries one, so its norm is at least one and the result follows. Suppose $K \geq 16$ and let r_x be the number of selected lines containing point x . If $r_x < K/4$ for every x , take the unit vector $a = \mathbf{1}/\sqrt{K}$. At each point,

$$|\sum_{j \in J} a_j h_j(x)| = |2r_x - K|/\sqrt{K} > \sqrt{K}/2.$$

Its Gram Rayleigh quotient is at least $K/4$. Otherwise choose x_0 incident with $r \geq K/4$ selected lines and let S be those r lines. At x_0 all these labels are positive; at any other point at most one is positive, since two points have a unique common line. For the unit vector $a = \mathbf{1}_S/\sqrt{r}$,

$$|\sum_j a_j h_j(x)| \geq (r - 2)/\sqrt{r} \geq \sqrt{r}/2 \quad \text{for all } x,$$

since $r \geq 4$. Its Rayleigh quotient is at least $r/4 \geq K/16$, for every μ . This proves (27), including marginals concentrated on a single point.

The claimed learner queries $q_j(x, y) = y h_j(x)$ for every line and outputs a hypothesis with maximum answer. The true target has correlation one, hence receives answer at least $1 - \tau$. A maximizing hypothesis has true correlation at least $1 - 2\tau$, and loss at most τ , regardless of the marginal and the oracle. Thus it is one distribution-independent learner.

We prove the fixed- D claim before excluding a common efficient learner. Let $\delta = \epsilon/2$ and sample a length- s sequence from the known D . For any pair of hypotheses at D -distance greater than δ , the probability of identical labels on the sequence is at most $(1 - \delta)^s \leq e^{-\delta s}$. A union bound over fewer than N^2 pairs is at most $N^2 e^{-\delta s} \leq e^{-1} < 1$. Consequently some sequence has the property that hypotheses with identical traces are within D -distance δ . Hardwire one such sequence and one representative hypothesis per trace into the learner. On at most s distinct sampled points there are at most $1 + s + \binom{s}{2}$ traces: the empty trace, at most s singleton traces, and at most one line through each pair of distinct sampled points. Every trace of size at least two is determined by one such pair. The representatives therefore form a D -cover of the claimed size. Query their label correlations and select the largest answer. A representative within δ of the true target has correlation at least $1 - 2\delta$. With tolerance $\tau = \epsilon/2$, the selected output has correlation at least $1 - 2\delta - 2\tau$ and error at most $\delta + \tau = \epsilon$. The hardwired cover depends on D ; this is not one learner serving all marginals.

To rule out a much cheaper common learner, choose for target line j the distribution

$$D_j(x) = \begin{cases} (2k)^{-1}, & P_{xj} = 1, \\ (2(N - k))^{-1}, & P_{xj} = 0. \end{cases}$$

Take the reference P_0 to be uniform on $X_q \times \{-1, +1\}$. With $c = k/(N - k)$, its residual likelihood ratios satisfy

$$r_j(x, +1) = A_{xj}, \quad r_j(x, -1) = -cA_{xj}.$$

For example the first formula is $2ND_j(x) - 1 = N/k - 1$ on the line and -1 elsewhere; the second is -1 on the line and $N/(N - k) - 1 = c$ elsewhere. Consequently

$$\begin{aligned}\Gamma &= \frac{1 + c^2}{2N} A^\top A, \\ \Lambda &= \frac{qN}{2} \left(\frac{1}{(q+1)^2} + \frac{1}{q^4} \right), \\ R &= \frac{2}{q((q+1)^{-2} + q^{-4})} \geq \frac{(q+1)^2}{q} \geq q.\end{aligned}$$

The penultimate inequality uses $q^4 \geq (q+1)^2$ for $q \geq 2$, which follows from $q^2 \geq q+1$. By Theorem 2, $R \leq 4 + 4M$ for $M = m/\tau^2$. Therefore

$$\text{dc}(\mathcal{H}_q) \leq 2q + 3 \leq 2R + 3 \leq 11 + 8M \leq 19M,$$

where the final inequality uses $M \geq 1$. □

Why projective planes do not separate. The family has unbounded dimension while all the proposed ordinary label-Gram ratios remain bounded. This is a genuine failure of the stated universality premise, not a counterexample to OQ2. The efficient learner needed for such a separation is impossible here: the target-dependent balanced marginals produce a joint-distribution ratio of order q . The lower-bound witness changes entry magnitudes, which the ordinary label Gram matrix cannot capture. This also verifies an explicit separation between fixed-distribution and distribution-independent SQ learning: $m_D = O(n^2/\epsilon^2)$ at tolerance $\epsilon/2$, whereas for $q \geq 16$ any common learner needs $m/\tau^2 \geq q/4$. Since $N \leq 2q^2$, the latter is at least $2^{(n-1)/2}/4$. The dimension lower bound is at least $2^{(n-1)/4}$. None of these fixed- D learners can be substituted for the single algorithm in OQ2.

A.3 Transcript compression

Theorem 5 (Marginal-branching and approximate probabilistic bounds). *Let $B = \lceil 2/\tau \rceil + 1$, with $0 < \tau \leq 1$. For a deterministic learner satisfying (1), decompose each query as*

$$q(x, y) = a(x) + yb(x), \quad a(x) = \frac{1}{2}(q(x, +1) + q(x, -1)), \quad b(x) = \frac{1}{2}(q(x, +1) - q(x, -1)).$$

Construct its marginal-response tree as follows. If a is constant, retain only the answer equal to that constant. Otherwise retain the B equally spaced answers in $[-1, 1]$. Let u be the maximum number of nonconstant- a nodes on a retained root-to-leaf path. Then

$$\text{dc}(\mathcal{H}) \leq (m+1)B^u. \tag{29}$$

In particular, a deterministic correlational-only learner has $\text{dc}(\mathcal{H}) \leq m+1$. For an arbitrary randomized learner satisfying (1),

$$\text{dc}^\epsilon(\mathcal{H}) \leq B^m. \tag{30}$$

Proof. The grid has spacing $2/(B-1) \leq \tau$, so nearest-grid rounding errs by at most $\tau/2$. The marginal tree has at most B^u leaves: suppress unary nodes and use induction on the number of remaining branching nodes per path. It has at most $(m+1)B^u$ nodes, because every node has a descendant leaf and a depth between zero and m , and assigning each node one such pair is injective when the depth is fixed along a leaf's path. Collect the function b at every internal node and the

Boolean output at every leaf. These are the common features $f_1, \dots, f_s$, where $s \leq (m+1)B^u$ and each feature has absolute value at most one.

Fix D, h . Follow the tree by giving the exact constant response or rounding $\mathbb{E}_D a$ to the grid. If at some node $|\mathbb{E}_D h b| > \tau/2$, one of the signed features $b, -b$ has correlation greater than $\tau/2$. Otherwise every answer is a valid SQ response: the difference from the true expectation $\mathbb{E}_D a + \mathbb{E}_D h b$ is at most $\tau/2 + \tau/2$. A valid policy realizing this finite transcript extends to all other histories by exact answers. The deterministic guarantee then gives leaf correlation at least $1 - 2\epsilon > 1/2 \geq \tau/2$. We have proved

$$\forall D, \max_{i, s_i \in \{-1, +1\}} \mathbb{E}_D[h(x)s_i f_i(x)] \geq \tau/2. \quad (31)$$

We supply the finite separation step. Let C be the convex hull of the vectors $(h(x)s_i f_i(x))_{x \in X}$ and let $Q = (\tau/2)\mathbf{1} + \mathbb{R}_{\geq 0}^X$. If $C \cap Q$ were empty, the compact set C and closed set Q would have a positive attained minimum distance. To see attainment, take a minimizing sequence; the C components have a convergent subsequence, and bounded distances make the Q components bounded too. A zero minimum would put the limit in both sets. Let (c, z) minimize the positive distance and put $v = z - c$. By differentiating squared distance along line segments, for every $c' \in C, z' \in Q$ we obtain

$$\langle v, c' - c \rangle \leq 0, \quad \langle v, z' - z \rangle \geq 0.$$

Taking $z' = z + t e_x$ gives $v_x \geq 0$. Furthermore $v \neq 0$, and

$$\max_{c' \in C} \langle v, c' \rangle \leq \langle v, c \rangle < \langle v, z \rangle = \min_{z' \in Q} \langle v, z' \rangle = (\tau/2) \sum_x v_x.$$

Normalizing v to a distribution contradicts (31). Hence a convex combination in $C \cap Q$ exists. Its coefficients define a linear combination of the common features whose product with $h(x)$ is at least $\tau/2$ at every point. This is a strict representation and proves (29). For correlational-only queries $a = 0$, hence $u = 0$.

For a randomized learner, fix a seed and instead retain all grid answers to each full query. There are at most B^m leaves. Use all leaf outputs as coordinates of an embedding, padding when needed. This distribution over embeddings depends only on the learner's seed, not on D, h . For fixed D, h , the policy rounding each true expectation to the grid is valid. Its output is one of the leaf features. Selecting that coordinate as w gives

$$\inf_w L_{D,h}(\text{sign}\langle w, \phi_\omega \rangle) \leq L_{D,h}(A_\omega^{O_{D,h}}).$$

Averaging over the seed proves (30). No seed is selected after observing D or h . $\square$

The deterministic argument does not derandomize the learner: fixing its seed need not preserve its expected guarantee. The probabilistic bound is exponential, not polynomial, and gives no exact $\text{dc}_{1/2}$ bound.

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